MICRO-DEGREE

Künstliche Intelligenz und Gesellschaft

Artificial Intelligence and Society

Micro-Degree Artificial Intelligence and Society

– Technical Aspects of AI –

Lecture 3.4: Algorithms for Unsupervised Learning

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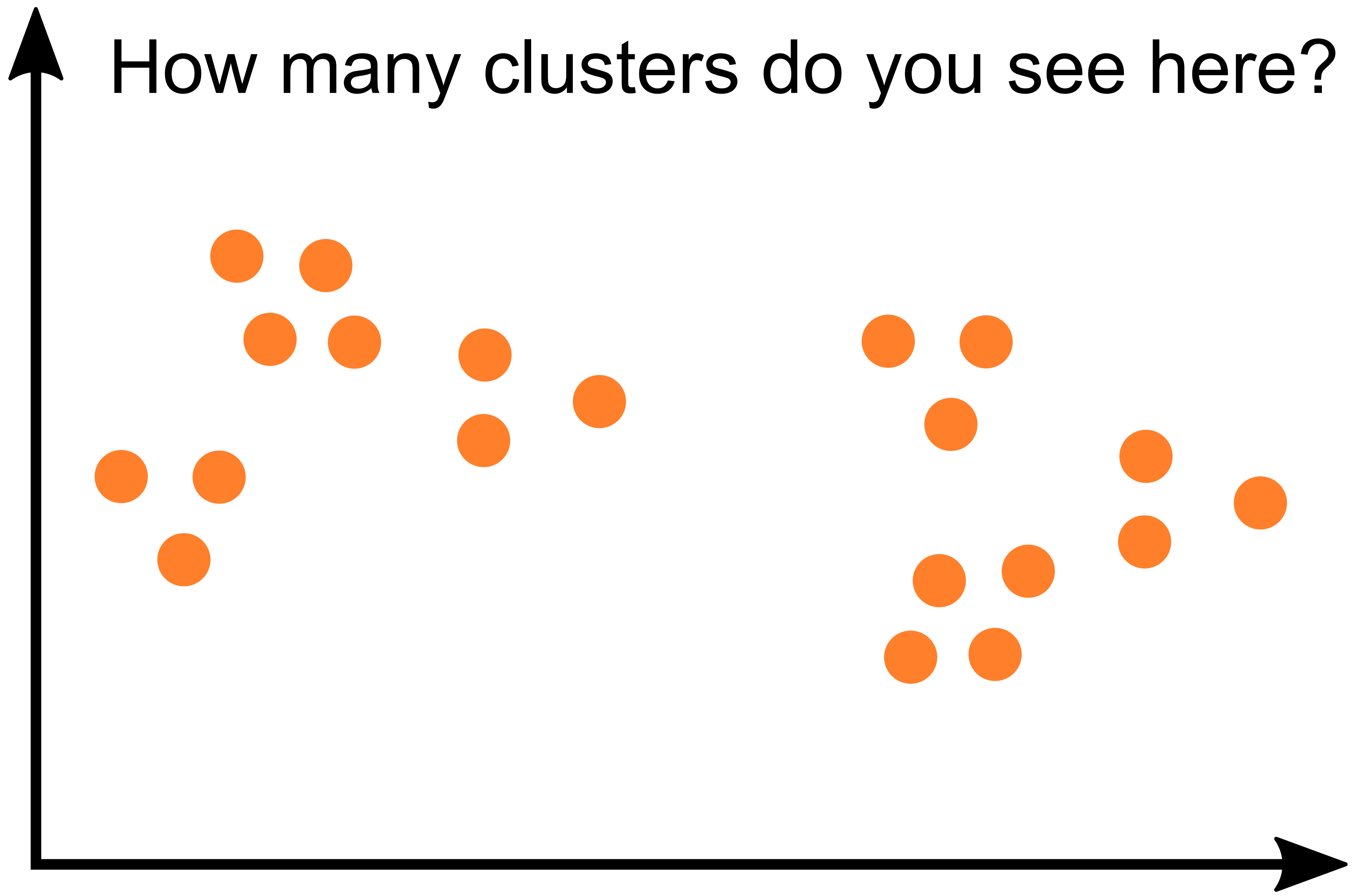
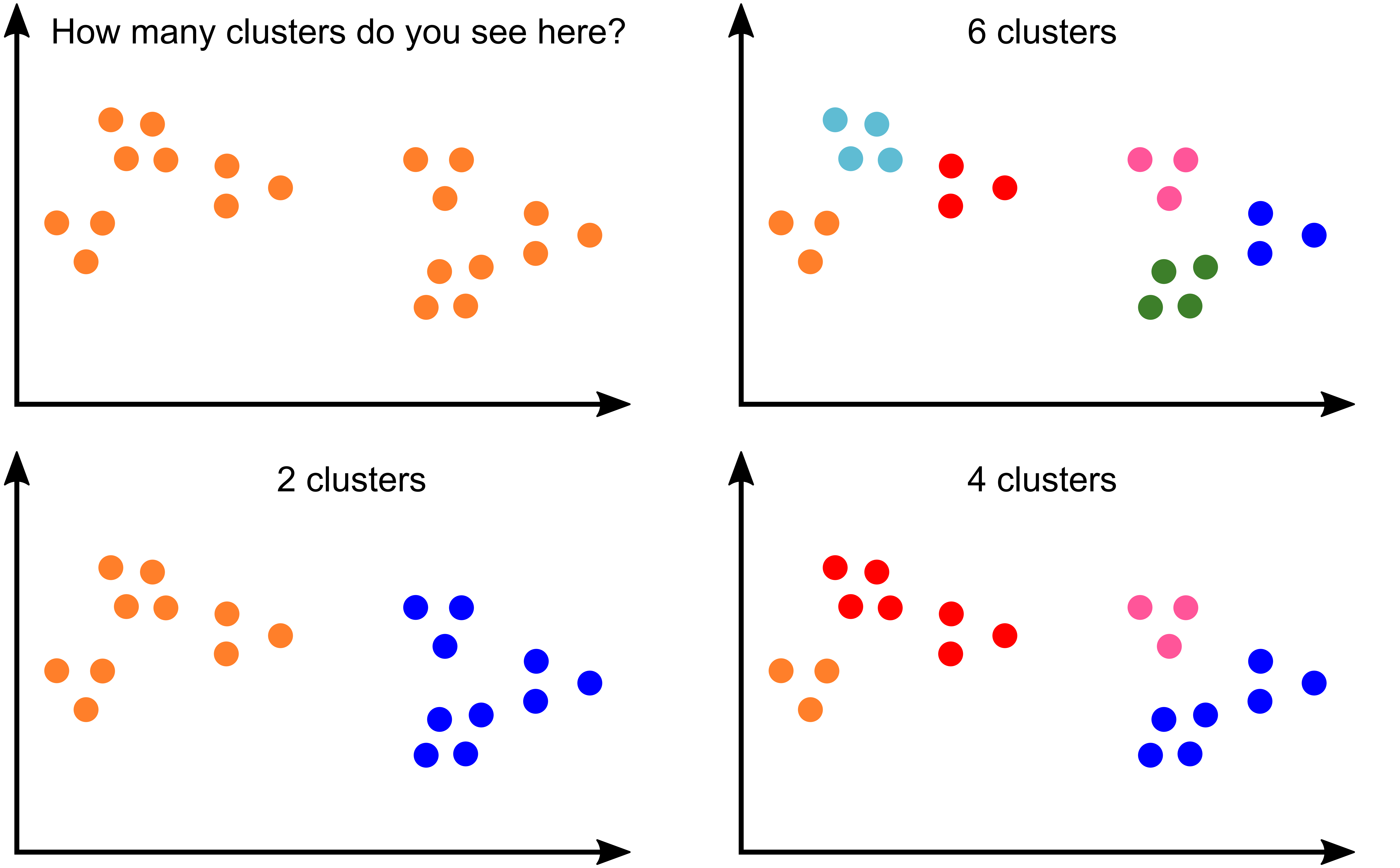
UNI GRAZ



Source: inspired by M. Hahsler, "Cluster Analysis: Basic Concepts and Algorithms".



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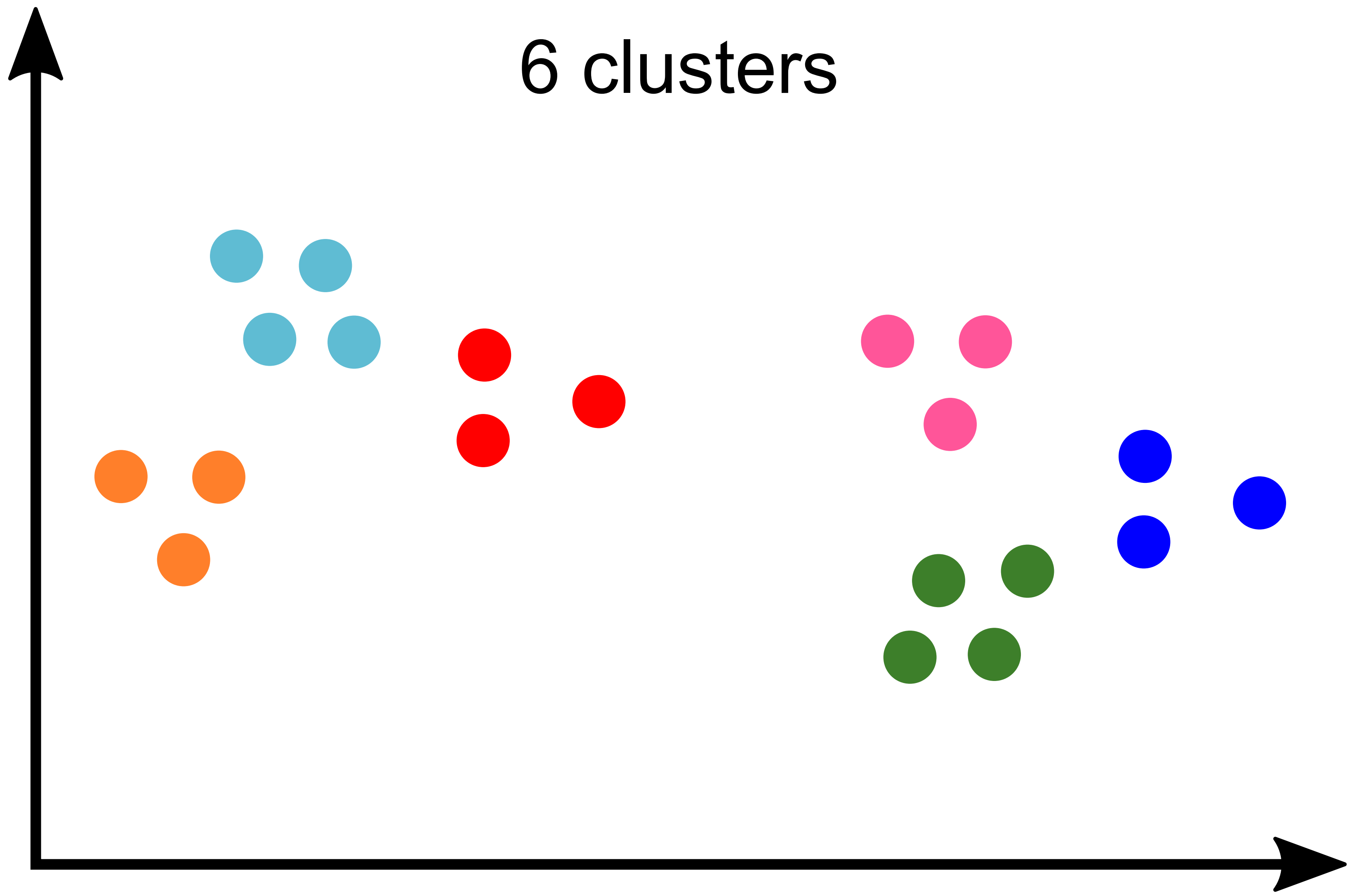
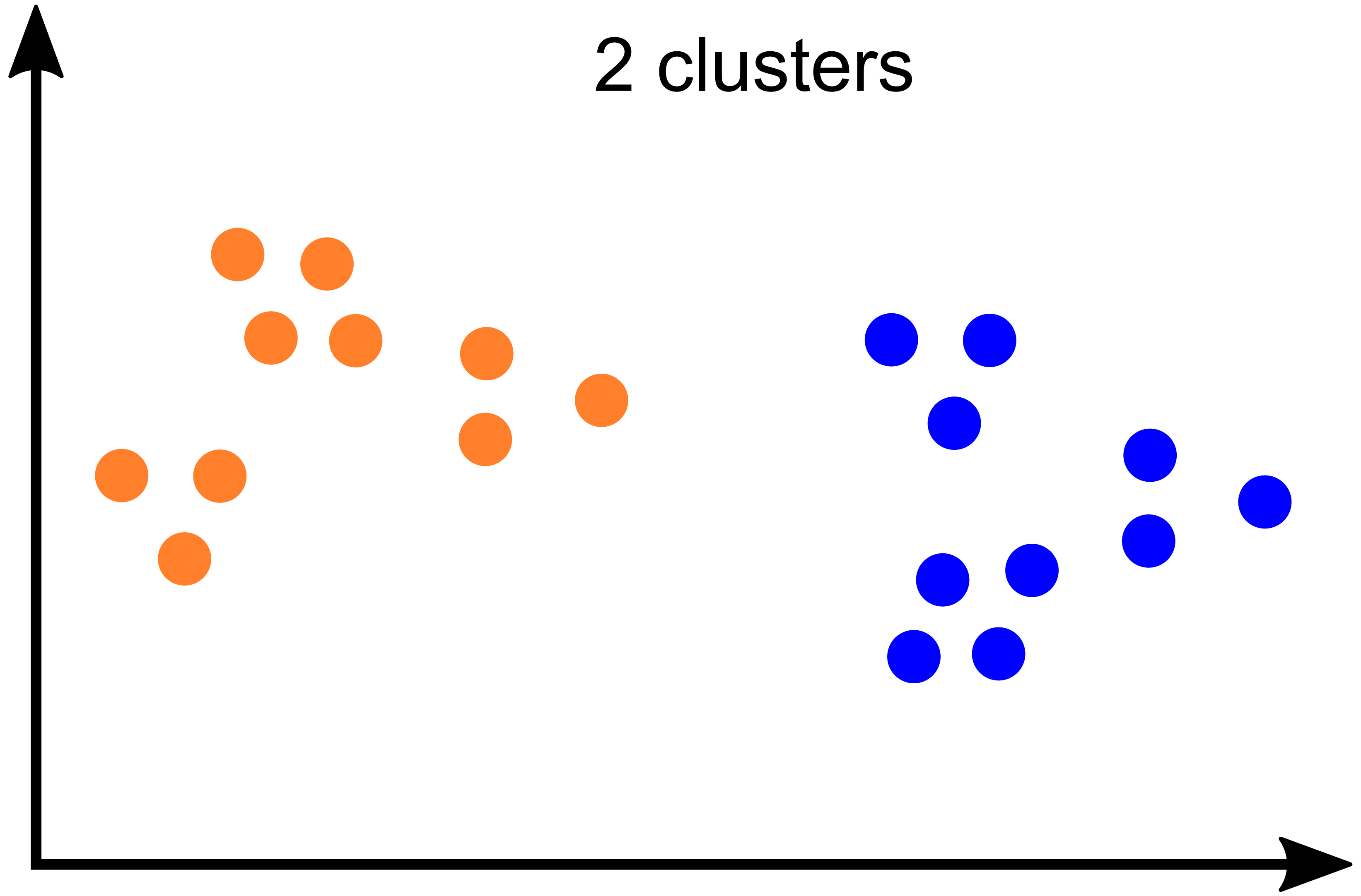
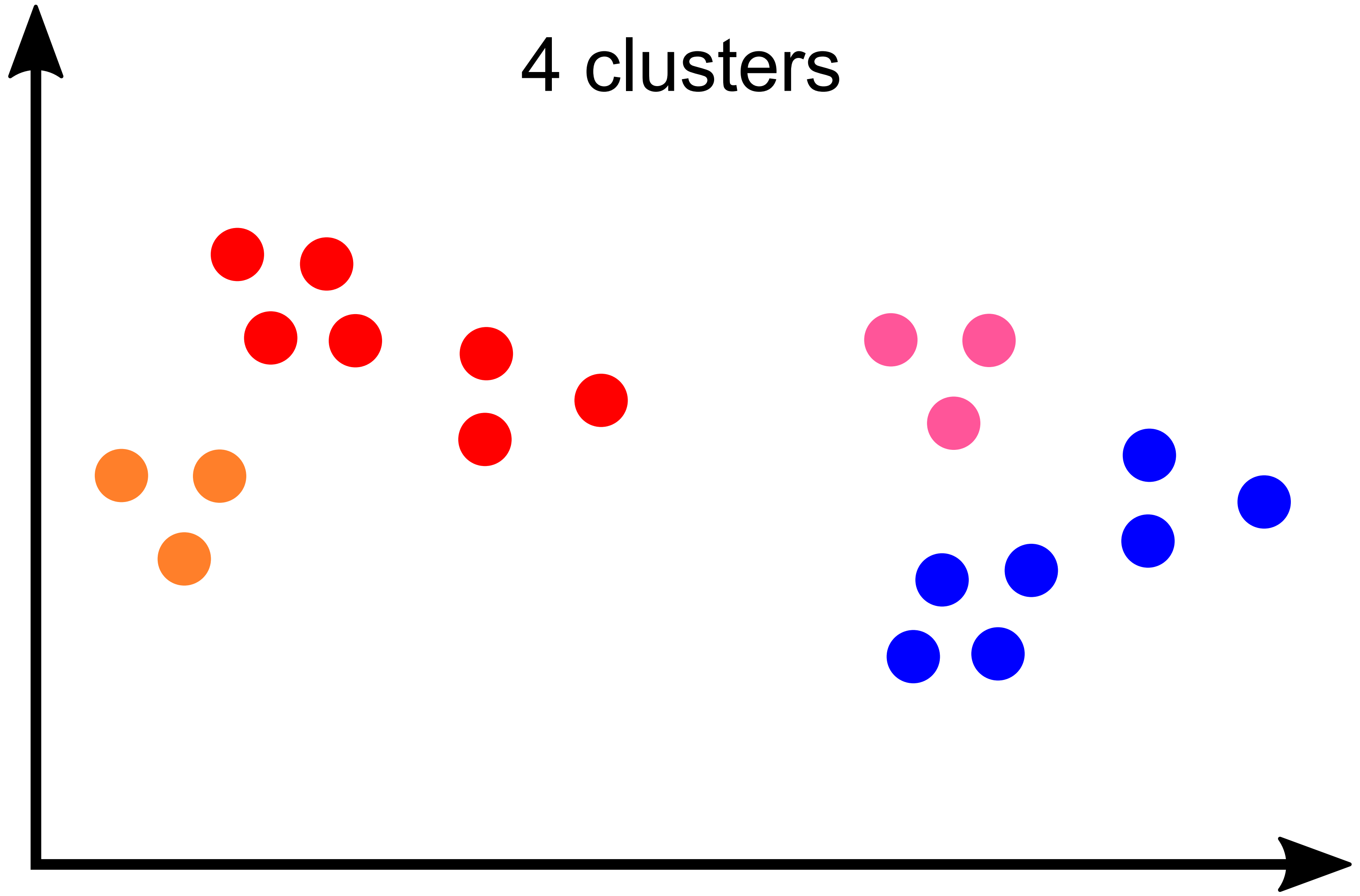
How to form clusters can be ambiguous!







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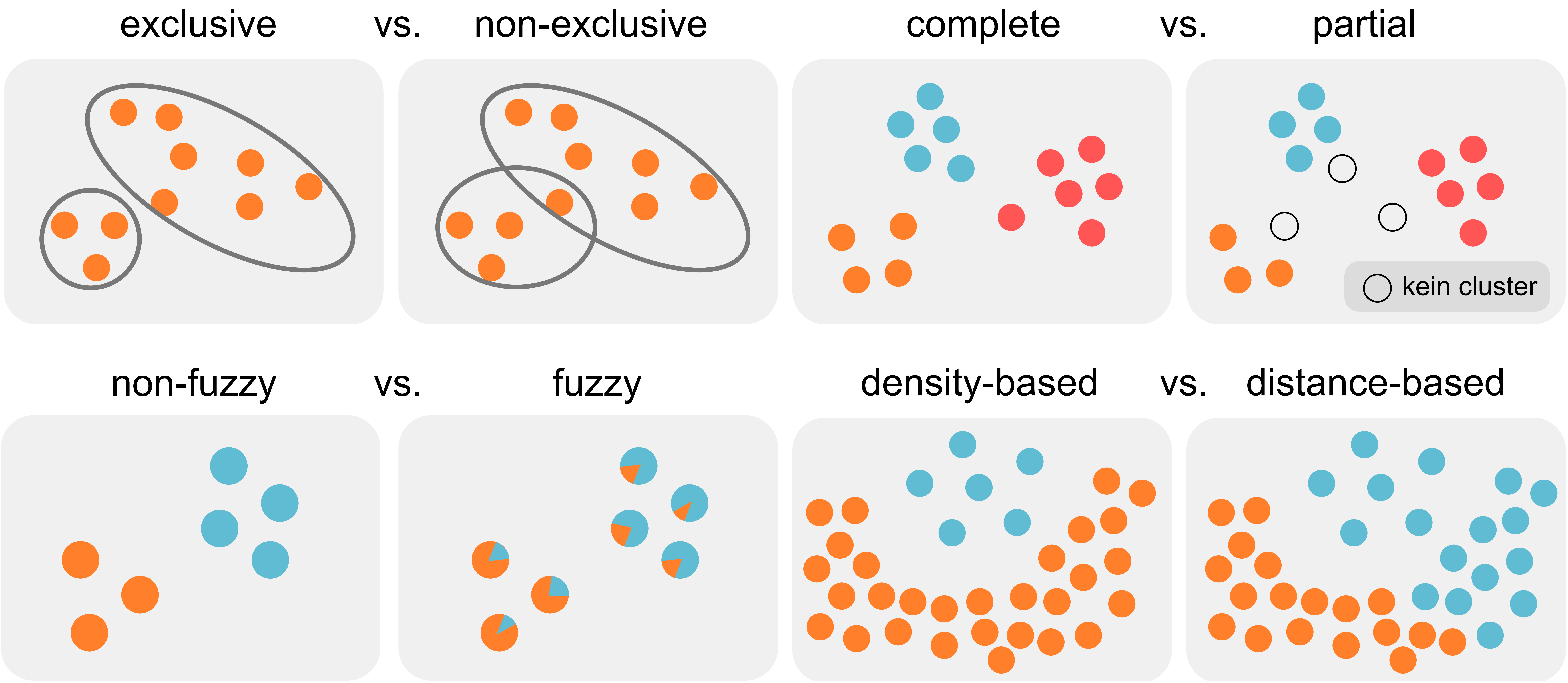
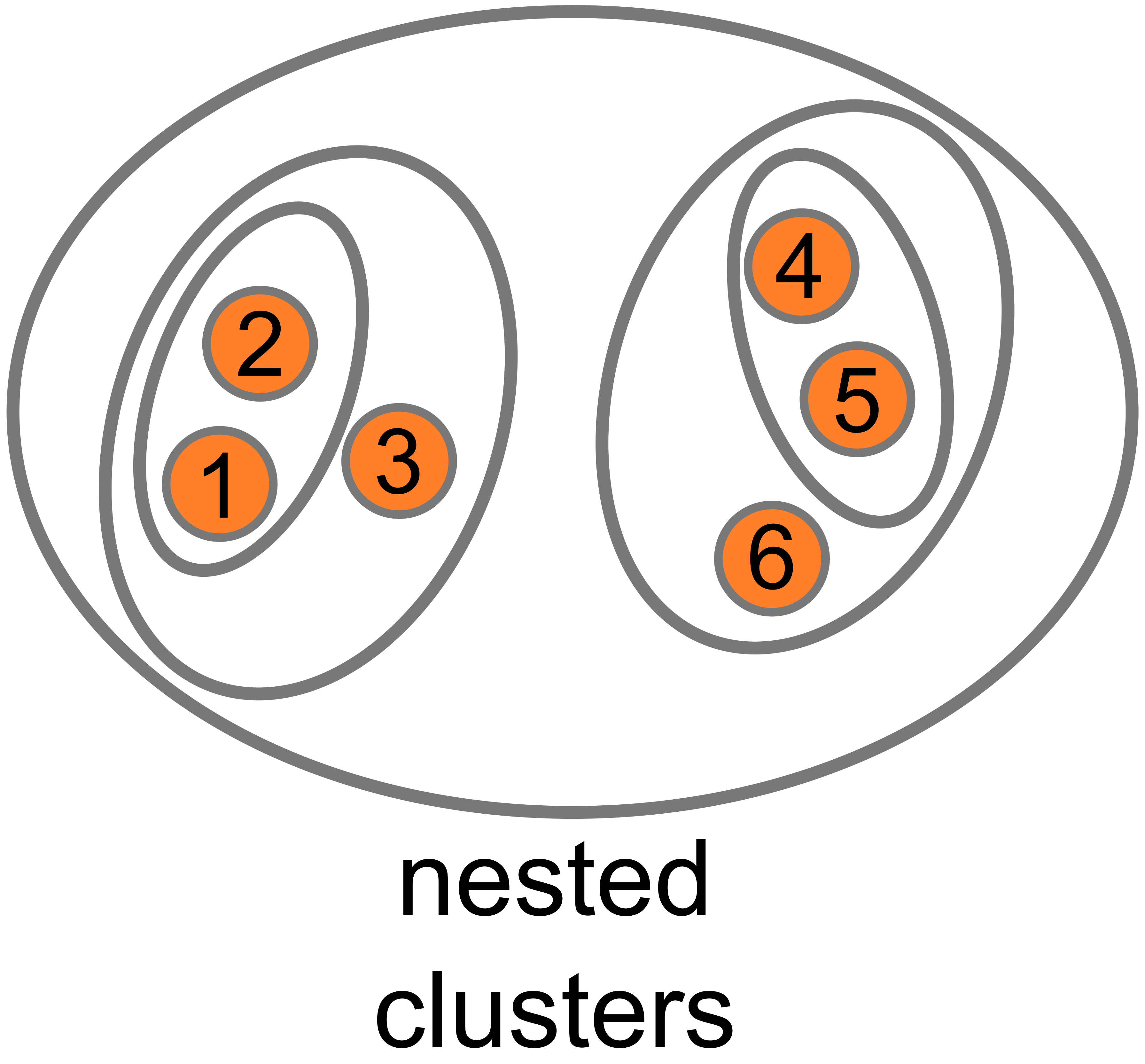
Different Clustering Properties

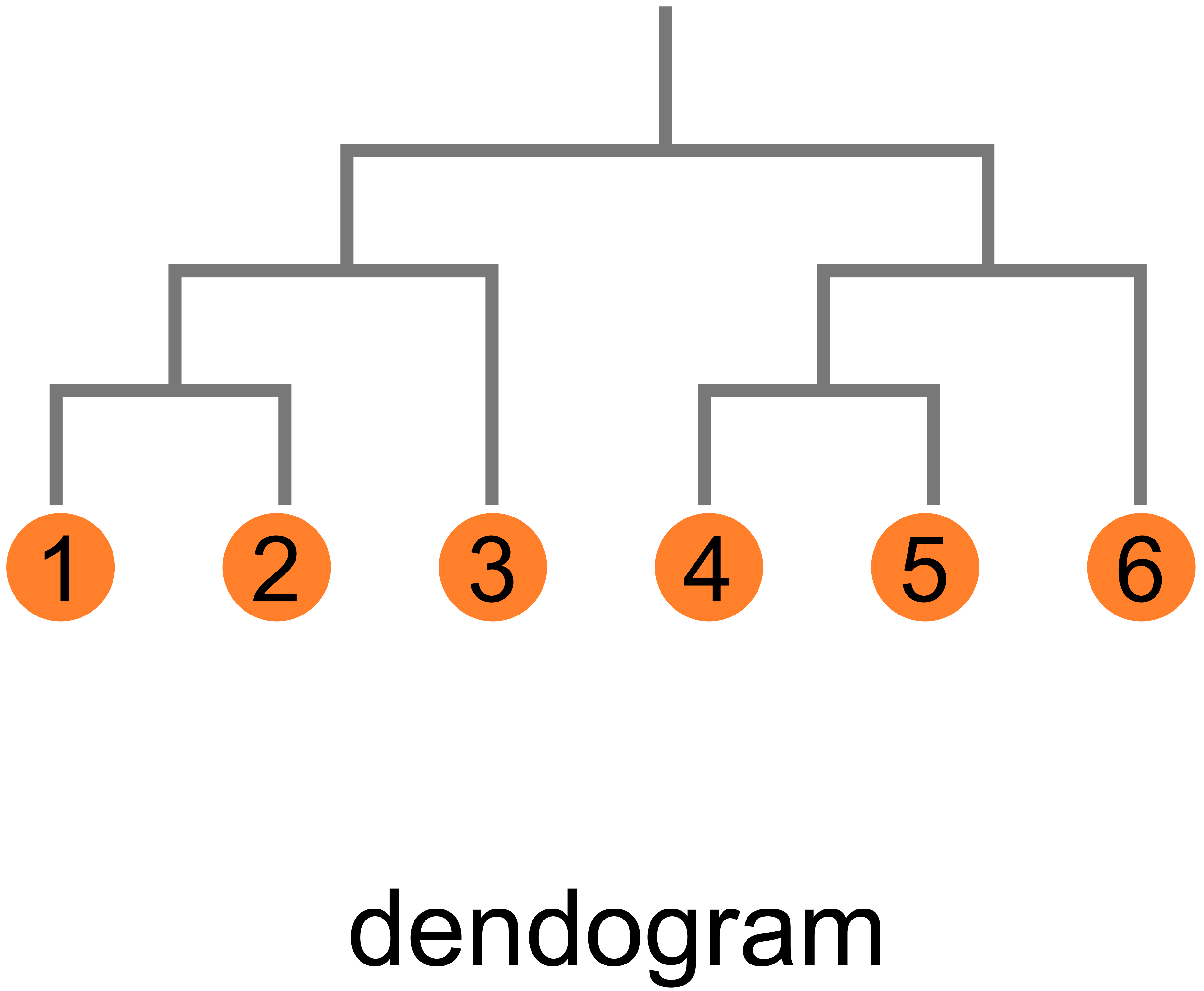


*Different clustering **methods** lead to clusterings with different properties.*

Hierarchical Clustering

Algorithm: *In each step, merge the two closest clusters.*





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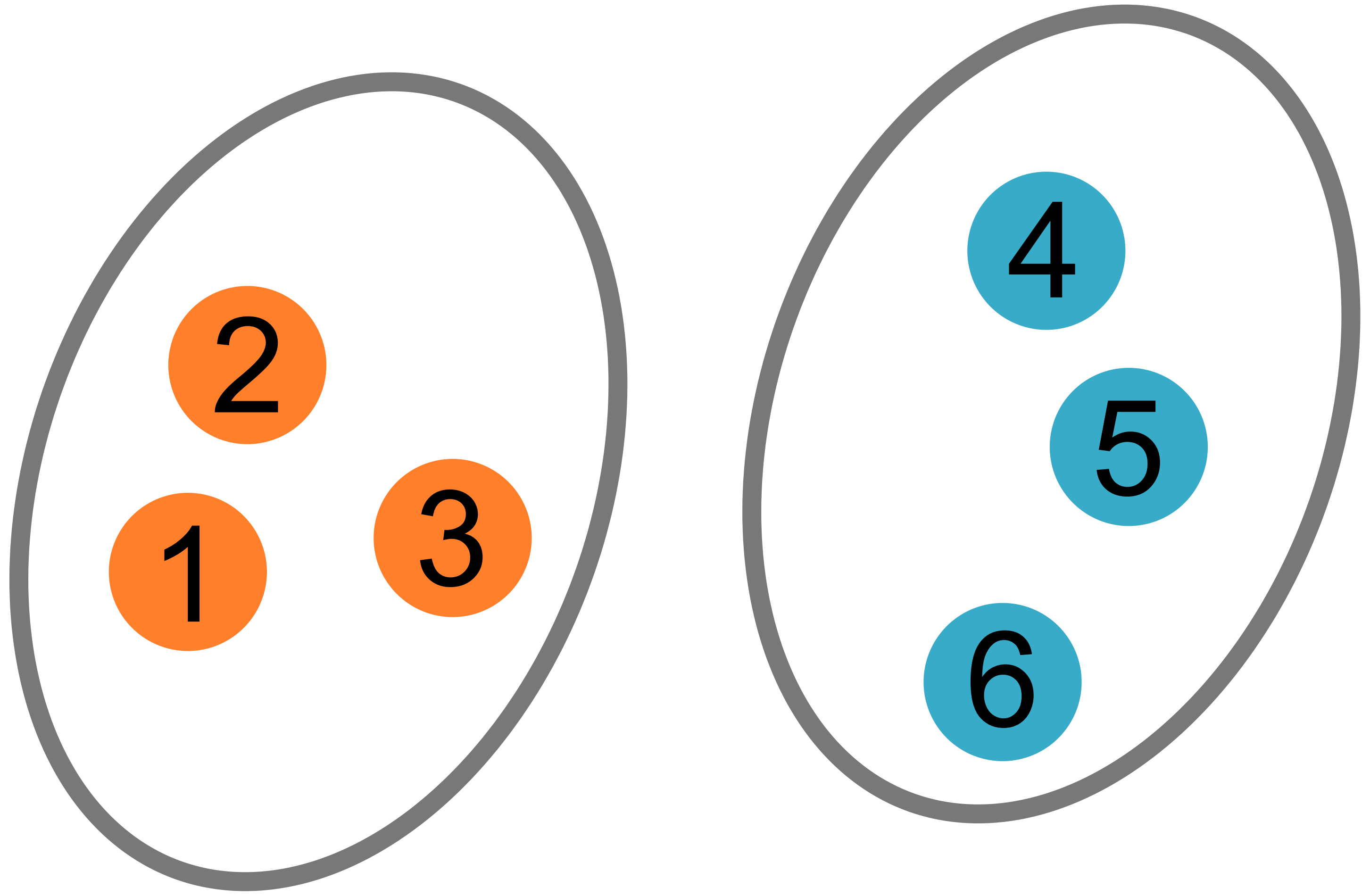
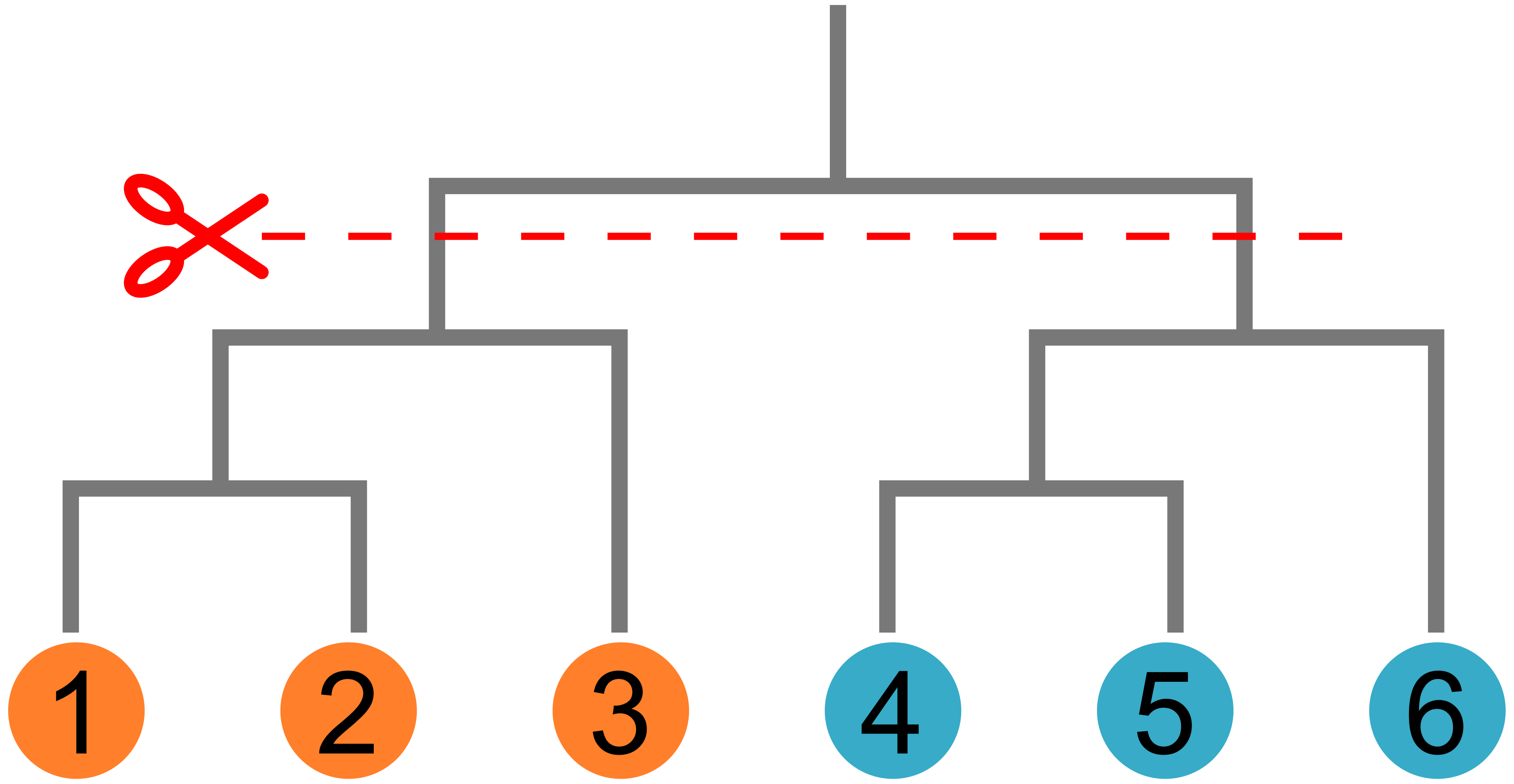


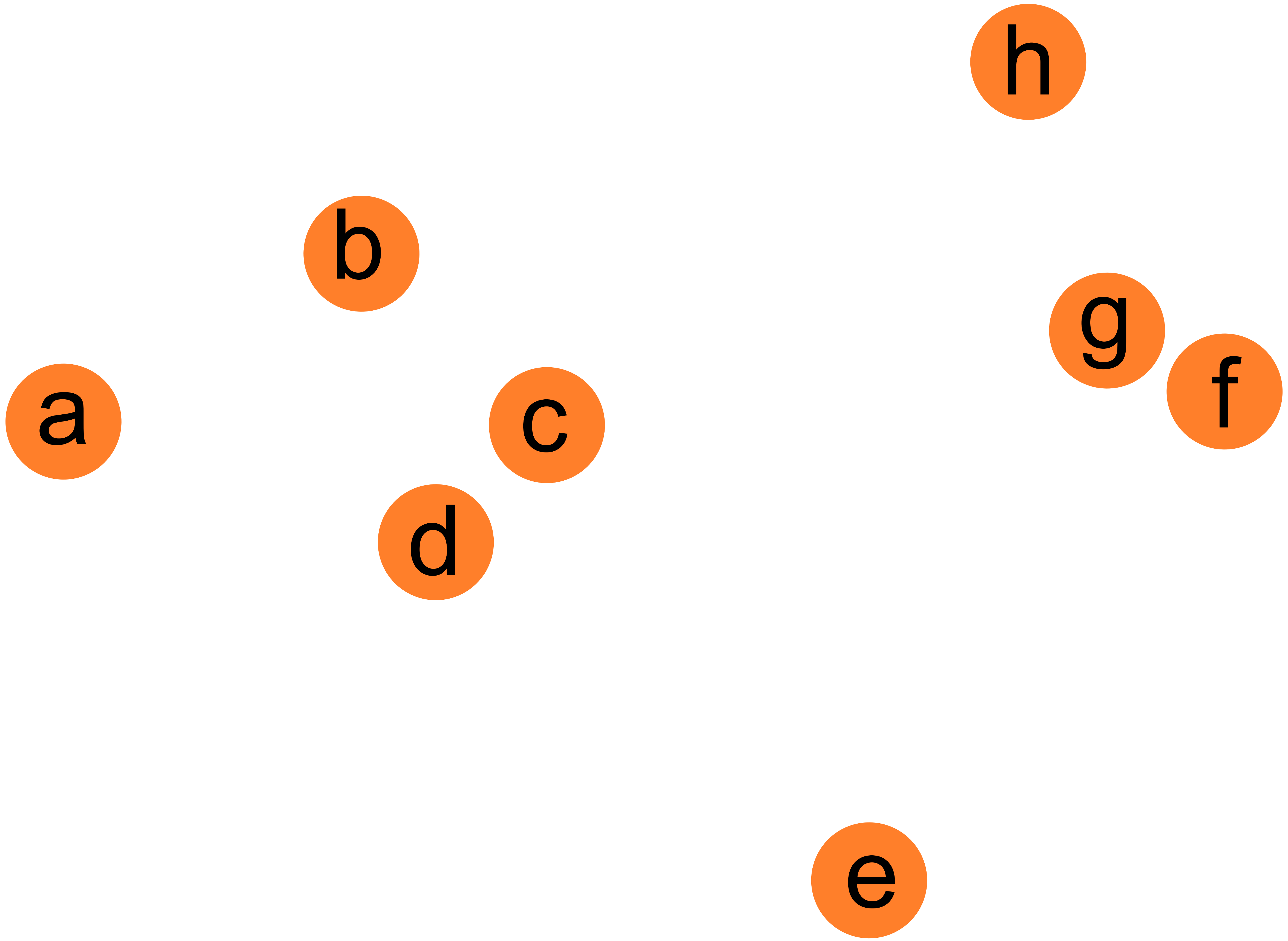


final clustering for
n clusters = 2

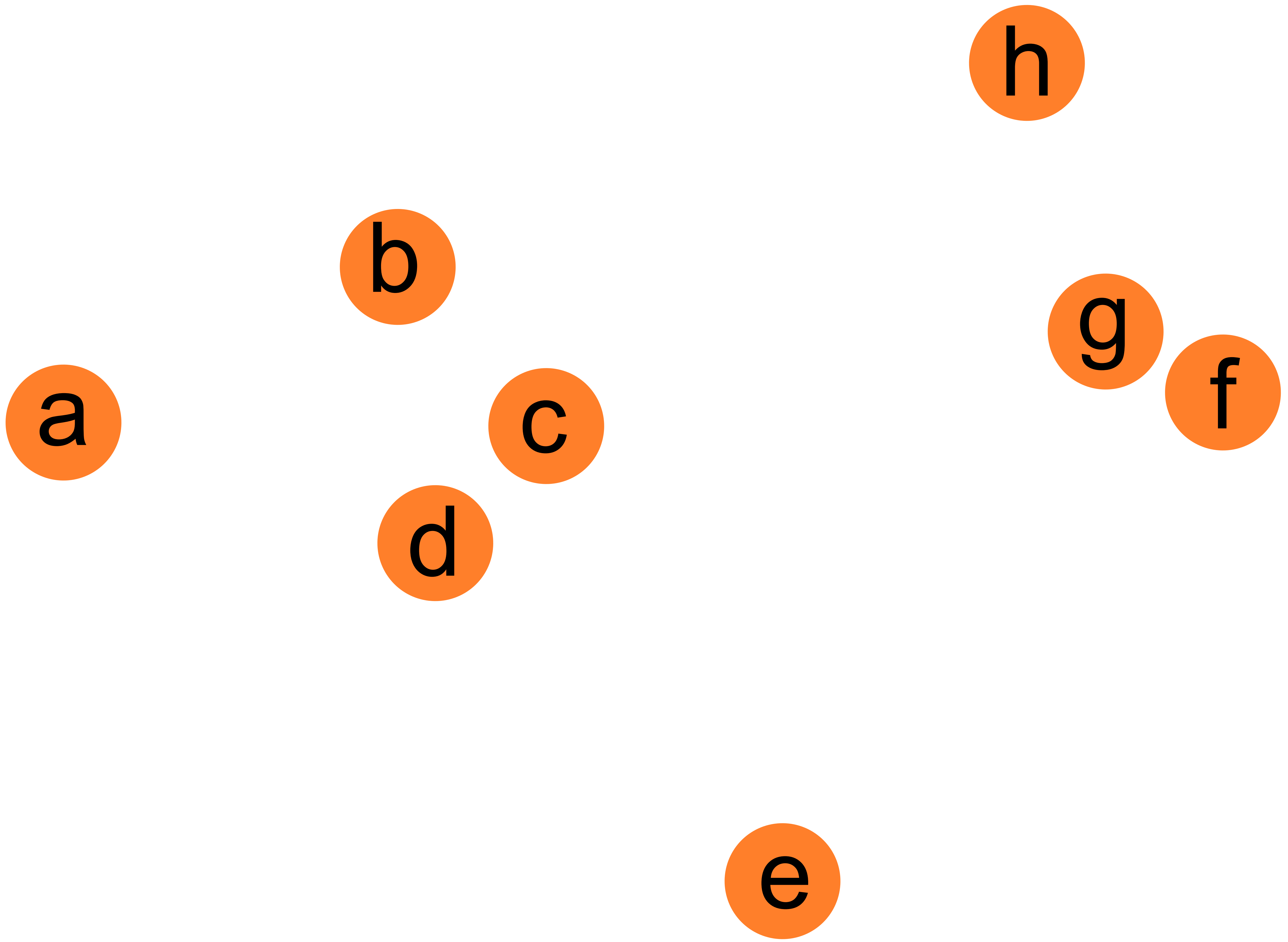
dendogram

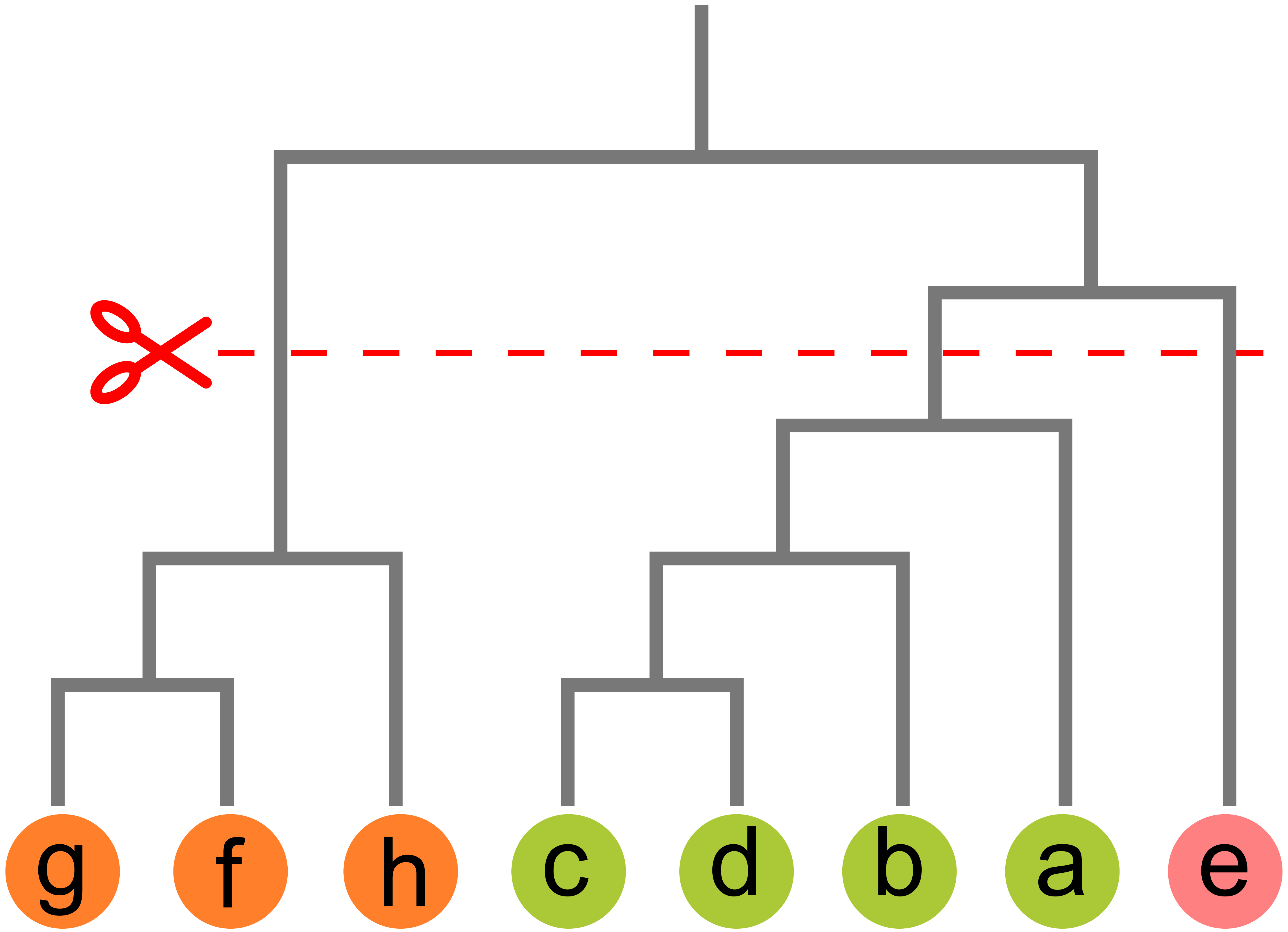
Exercise (5 min)

Perform a hierarchical clustering for the depicted data points based on their Euclidean distance and draw a dendrogram. What grouping do you get for n=3 clusters?



Solution





Partitioning Clustering Methods

*Starting from a given clustering, move data points to other clusters to find a **better** clustering.*

Objective Functions: Intuition

*To find the **best** clustering, the algorithm optimizes its **objective function**.*

*Example: **"Sum of Errors"**.*

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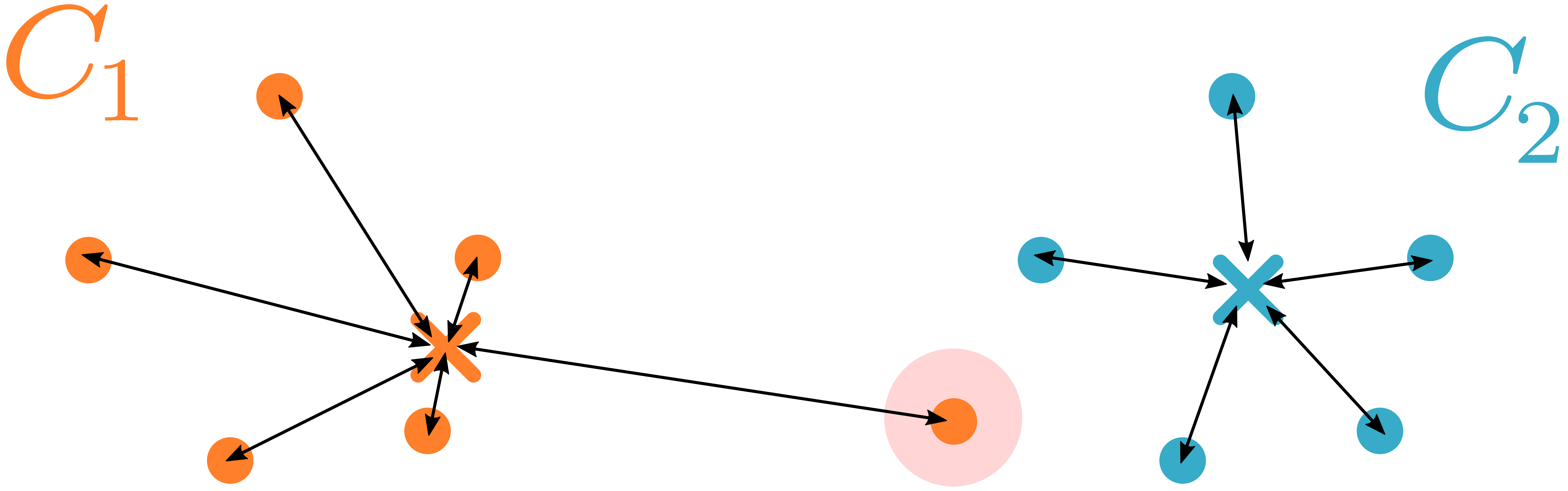

Objective Functions: Intuition

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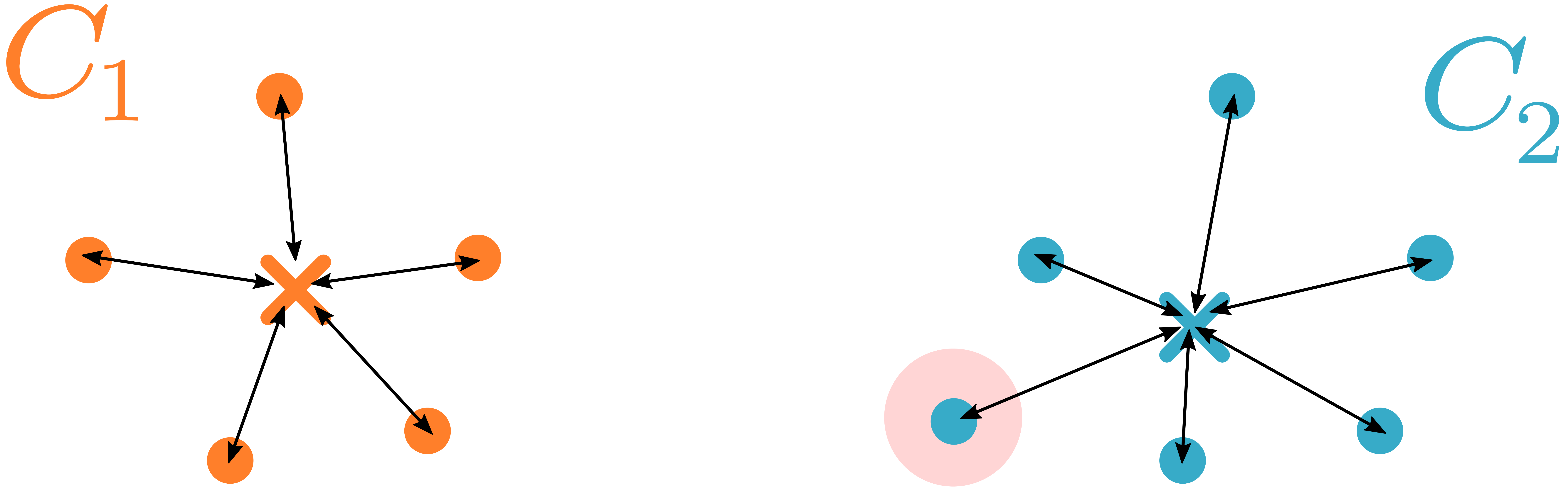
*Example: **"Sum of Errors"**.*



Sum of Squared Errors over the Clustering

The sum of squared errors is also called "Sum of Squared Errors" (SSE).

sum over all data points *J* in cluster C_i

value of attribute *k for* data point *j* in cluster *i*.

$$SSE = \sum_{i=1}^{N} \sum_{j=1}^{J_{C_i}} \sum_{k=1}^{K} (x_{ijk} - \bar{x}_{ik})^2$$

sum over all clusters *N*

sum over all attributes *K*

mean value of attribute *k* in cluster *i*

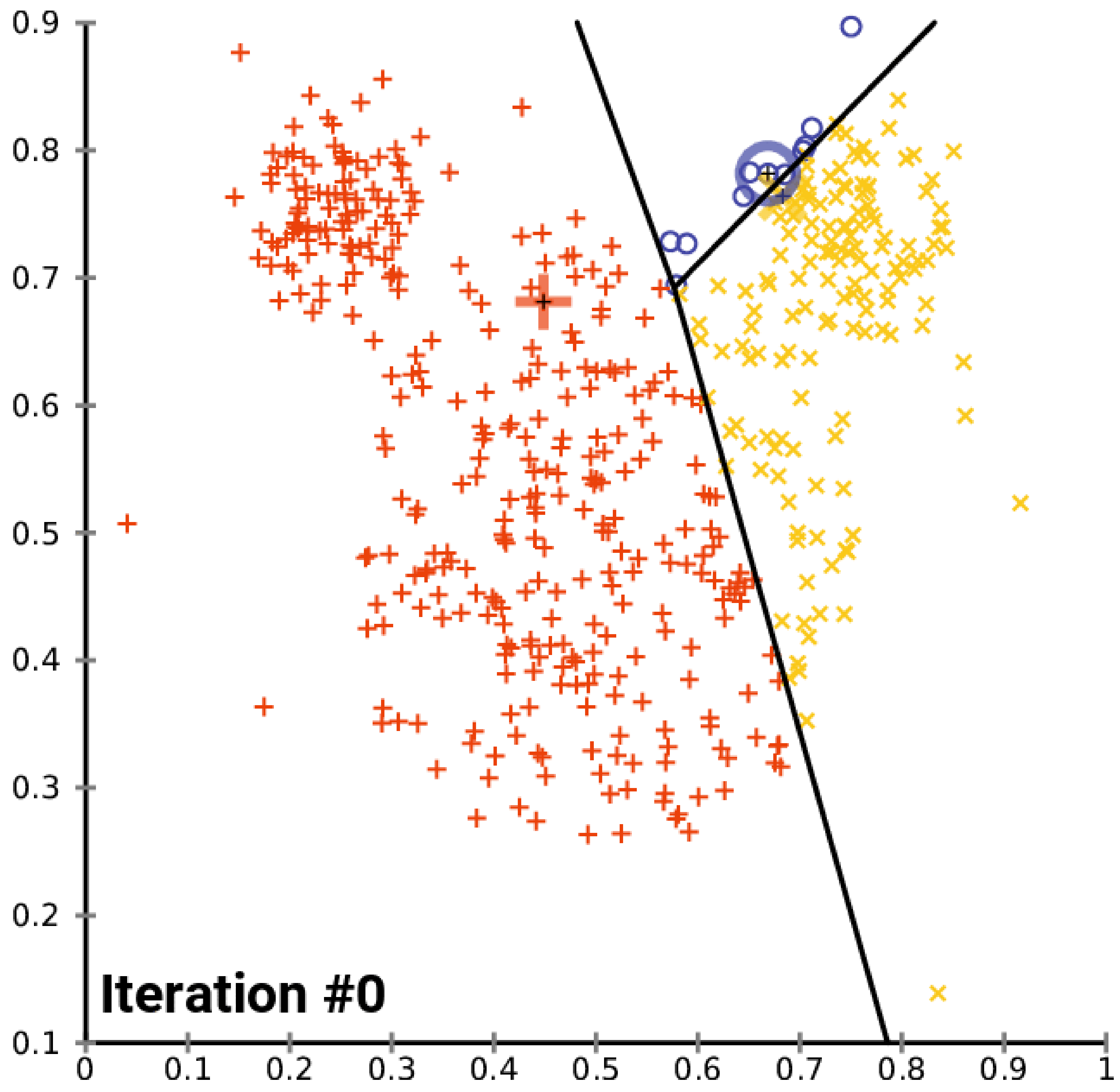
K-means:
an algorithm for finding clusters

Intuition: Starting from a given clustering, move data points to other clusters to ***reduce the Sum of Squared Errors (SSE)****.*

K-means Algorithm

1. Propose a clustering.
2. Calculate the mean per attribute & cluster.
3. Calculate *SSE* over the entire clustering.
4. For all data points: check if *SSE* can be reduced by changing cluster membership.
5. Move the data point with the maximum reduction of SSE to the corresponding cluster.
6. Calculate new means of all attributes for the giving and receiving cluster.

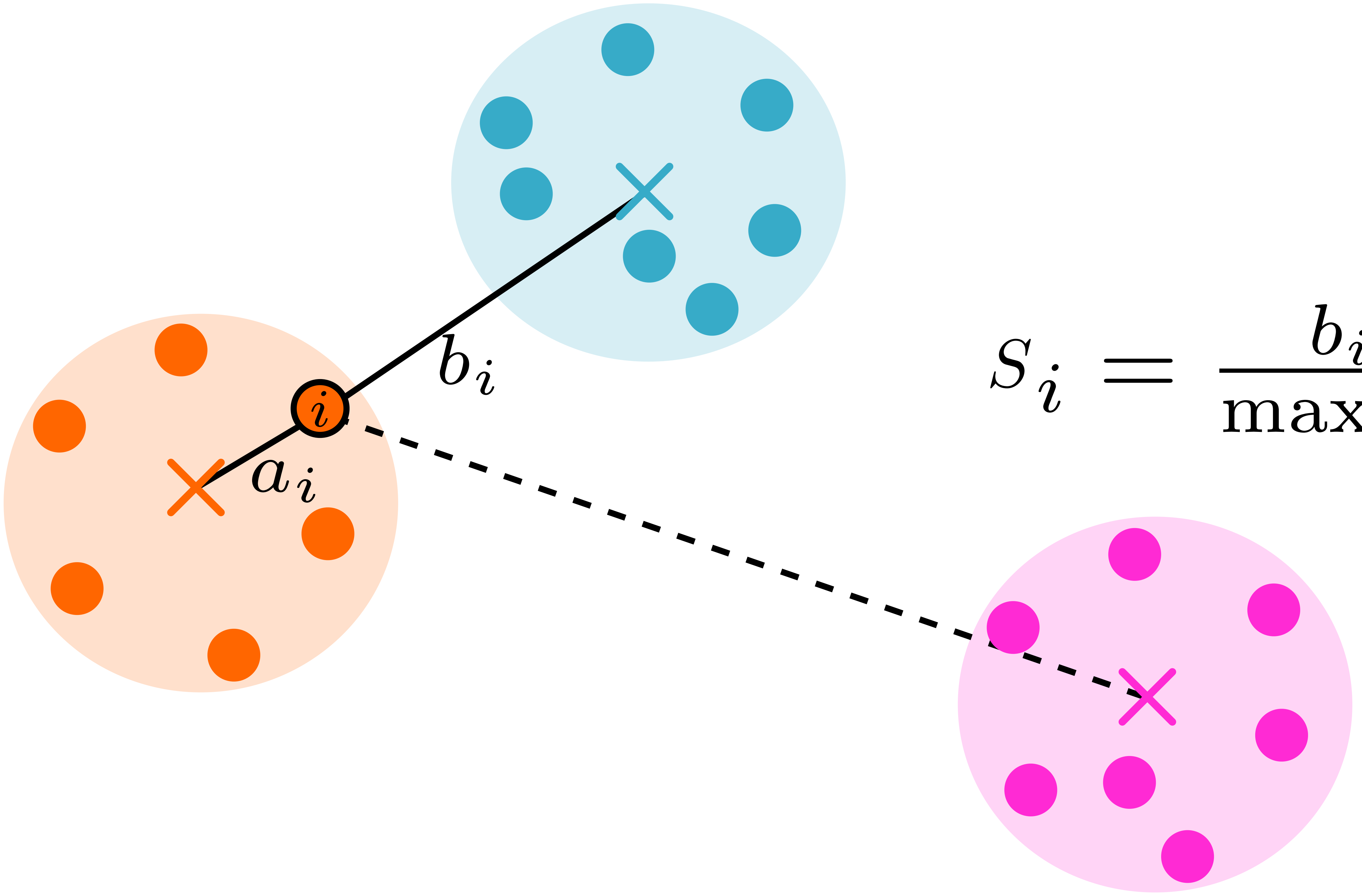
Repeat steps 4 to 6 until SSE doesn't go down anymore.



What is the Optimal Number of Clusters?

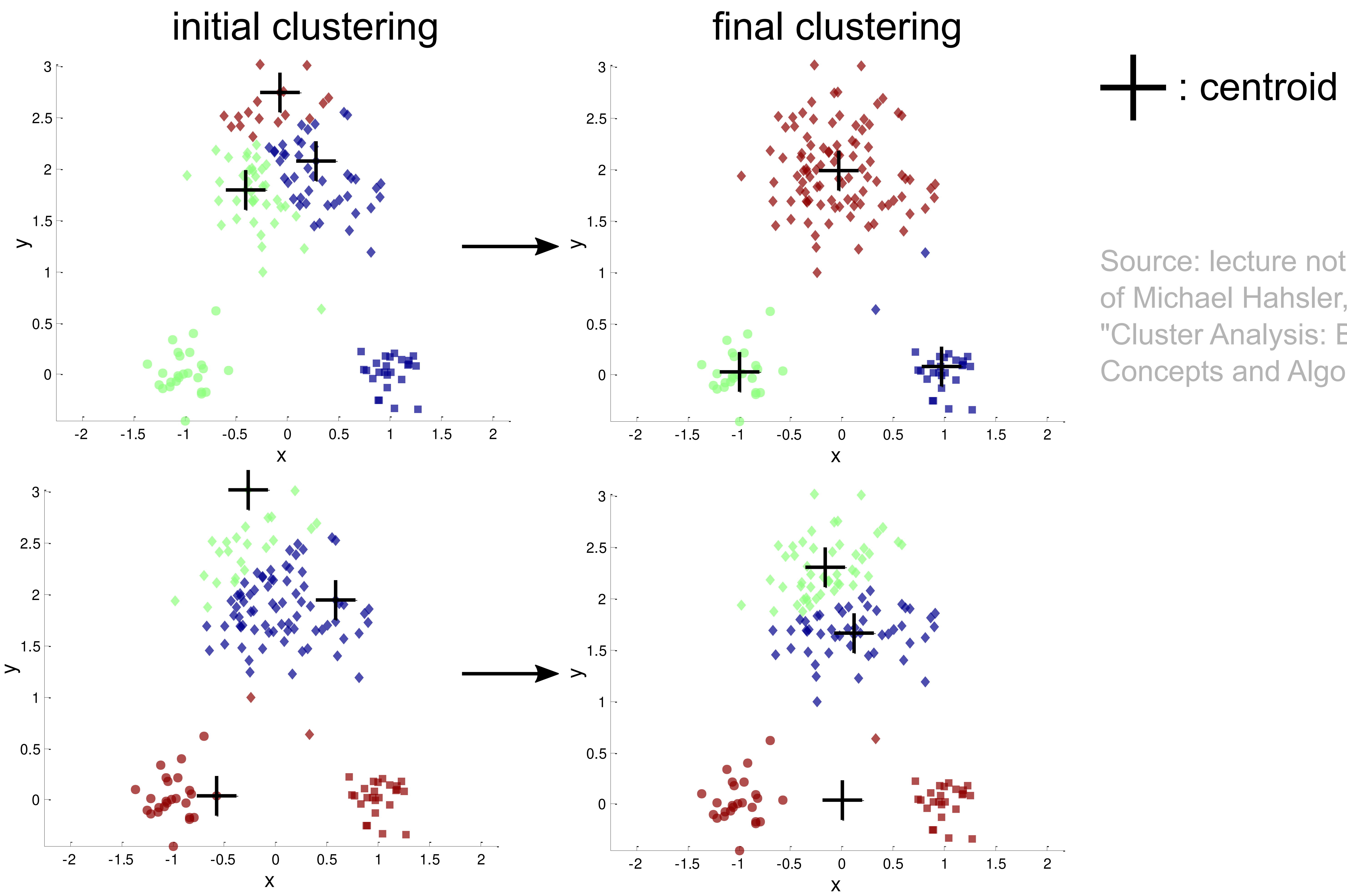
Silhouette Score: measures how similar an observation is to other observations in the same cluster (cohesion) compared to observations in other clusters (separation). Here, **1 is very good** and **-1 is very bad**.

- ***a_i***: mean distance of observation *i* to all other observations in the same cluster.
- ***b_i***: mean distance to the next closest cluster.
- The silhouette score is the mean of s_i over all observations.



$$s_i = \frac{b_i - a_i}{\max(a_i, b_i)}$$

Influence of the Initial Clustering



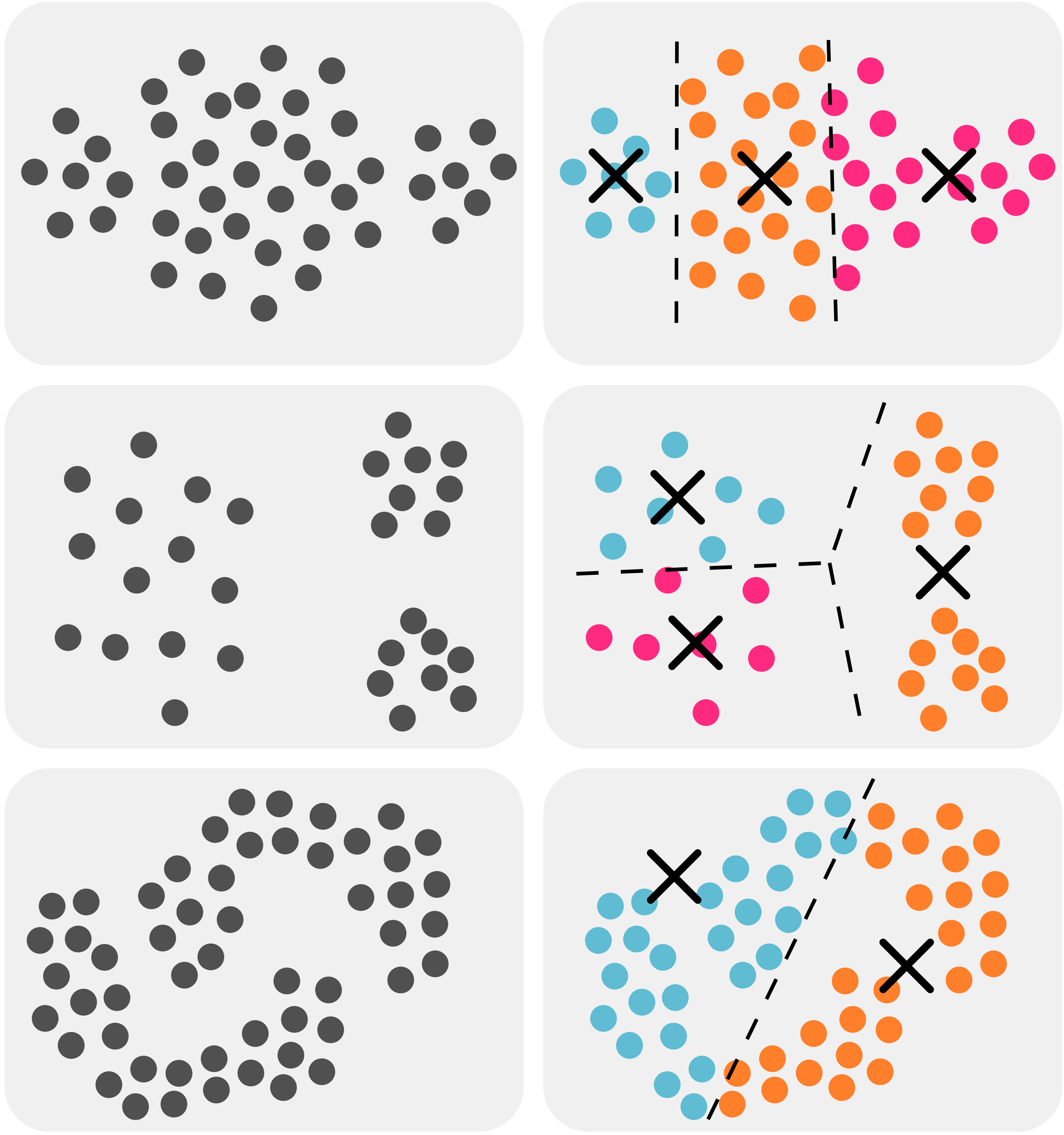
Source: lecture notes of Michael Hahsler, "Cluster Analysis: Basic Concepts and Algorithms"

→ Repeat the clustering procedure several times with different initial clusterings.

Limitations of K-Means Clustering

K-means leads to unintuitive results when the clusters are ...

- ... of different sizes.
- ... of different densities.
- ... do not have a convex hull.



→ Density-based clustering (see for example DBSCAN)

Summary

- Different clustering methods produce different clusterings
  - Hierarchical methods
  - Partitioning methods (k-Means)

- Clusterings can have **different properties.**
  - fine vs. coarse-grained
  - exclusive vs. non-exclusive
  - complete vs. partial
  - density-based vs. distance-based